

Data and Background

Background

Medicines are an important part of healthcare, and examining pharmaceutical dispensing data can help us understand how medicine use varies across New Zealand, between geographic areas and over time. Dispensing patterns may be influenced by differences in population size, age structure, socioeconomic conditions and healthcare needs. This project uses pharmaceutical dispensing data to explore these patterns, with a particular focus on medicines associated with dementia.

Dementia describes a group of conditions that cause a progressive decline in cognitive functions such as memory, thinking and the ability to carry out everyday activities. It is more common among older people and can create substantial healthcare and support needs. Understanding patterns in dementia-related pharmaceutical dispensing is therefore important, particularly as differences may exist between geographic areas and population groups.

The group's analysis focuses on Donepezil hydrochloride and Rivastigmine, two medicines used in the management of symptoms associated with dementia. We examine how dispensing of these medicines changed between 2020 and 2024 and how patterns differed between New Zealand health districts. However, raw dispensing counts alone may not provide a fair comparison between districts. For example, an area with a larger population aged 65 years and over may naturally have more dementia-related medicine dispensings. Population and socioeconomic deprivation information were therefore incorporated to provide additional context when examining geographic differences.

The pharmaceutical data were obtained from Te Whatu Ora – Health New Zealand and contain aggregated information about medicines dispensed through community pharmacies in New Zealand. The initial exploratory analysis examines general pharmaceutical patterns, including commonly dispensed medicines, changes over time, geographic variation and therapeutic groups, before focusing more closely on Donepezil and Rivastigmine.

Data

Three main sources of data were used: pharmaceutical dispensing data, socioeconomic deprivation data, and population data. The pharmaceutical dispensing data were obtained from Health New Zealand | Te Whatu Ora's Pharmaceutical Data Web Tool. The socioeconomic deprivation data and population estimates were obtained from

[\[\[https://explore.data.stats.govt.nz/vis?pg=0&bp=true&snb=19&tm=deprivation%20index&hc\\[N\]\(https://explore.data.stats.govt.nz/vis?pg=0&bp=true&snb=19&tm=deprivation%20index&hc\[N\)](https://explore.data.stats.govt.nz/vis?pg=0&bp=true&snb=19&tm=deprivation%20index&hc[N)

[ew%20Zealand%20index%20of%20socioeconomic%20deprivation\]=&df\[ds\]=ds-nsiws-disseminate&df\[id\]=CEN23_DEP_019&df\[ag\]=STATSNZ&df\[vs\]=1.0&dq=2023.99999..9999.99.99&to\[TIME\]=false](#)]. These datasets have different levels of granularity and were integrated where required for the dementia analysis.

The pharmaceutical data were provided in several CSV files, including Data_ByChemical.csv, Data_ByTG2.csv, Data_ByTG3.csv and PharmaceuticalsLookup.csv. The chemical-level dataset was the main dataset used for analysing individual medicines. Each row represents an aggregated combination of characteristics such as pharmaceutical chemical, health district, year and dispensing type, rather than an individual patient. The TG2 and TG3 datasets contain more aggregated therapeutic-group and subgroup information, while the lookup table connects individual pharmaceutical chemicals to their corresponding therapeutic classifications.

The socioeconomic deprivation data provide information on deprivation across geographic areas. For the analysis, these data were used to calculate a population-weighted deprivation measure for each health district, providing district-level socioeconomic context. The population data contain population estimates by geographic area, year and age group. For the dementia analysis, these were aggregated to obtain estimates of the population aged 65 years and over for each district and year. This allowed dementia-related dispensing patterns to be considered relative to the size of the older population rather than comparing districts using raw dispensing counts alone.

Table 1. Main variables in the chemical-level pharmaceutical dataset

Variable	R data type	Description
Chemical	Character	Name of the pharmaceutical chemical
ChemID	Character	Identifier associated with the pharmaceutical chemical
District	Character	Health district or New Zealand national aggregate
YearDisp	Numerical	Year in which the dispensing was recorded
NumDisps	Numerical after cleaning	Number of pharmaceutical dispensings
NumPpl	Numerical after cleaning	Number of people associated with the dispensing record

Although variables such as Chemical, District and Type are stored as character variables in R, they can be treated statistically as categorical variables. NumDisps and NumPpl were initially stored as character variables because some observations contain the value <6. After treatment of these suppressed observations, they were converted to numerical variables.

Data quality and preparation

Before the main analysis, the pharmaceutical data were checked for missing values, duplicate records and variable structure. Records were then filtered to include Type = "Dispensings", excluding initial dispensing records before suppression was assessed. After this filtering, 4127 records had suppressed NumDisps values and 10147 records had suppressed NumPpl values.

Suppression occurs when a small count is reported as <6 rather than providing its exact value. For numerical analysis, these values were replaced with 3 for both NumDisps and NumPpl, allowing the observations to remain in the analysis rather than being removed. However, the actual suppressed values are unknown, so imputing a value of 3 introduces a small amount of uncertainty into calculated totals, rates and subsequent analyses.

Integration of the three data sources also required attention to the join keys. District names were not completely consistent between datasets because of differences in naming conventions and the inclusion of Māori macrons. For example, Tairawhiti and Waitemata in the pharmaceutical data were standardised to Tairawhiti and Waitemata. Capital & Coast and Hutt Valley were also combined as Capital, Coast and Hutt Valley where required to match the geographic structure of the other data. Standardising these names was necessary because inconsistent join keys could result in records for the same geographic area failing to match.

After standardisation, pharmaceutical and deprivation information could be linked by district, while population information was linked using district and year. These integrated data provide the basis for examining whether differences in Donepezil and Rivastigmine dispensing across New Zealand are associated with geographic, demographic and socioeconomic characteristics.